8 **(a)** Fig. 8.1 shows the variation of the gravitational potential ϕ with distance d from the surface of Pluto along a line joining the centers of Pluto and Charon.

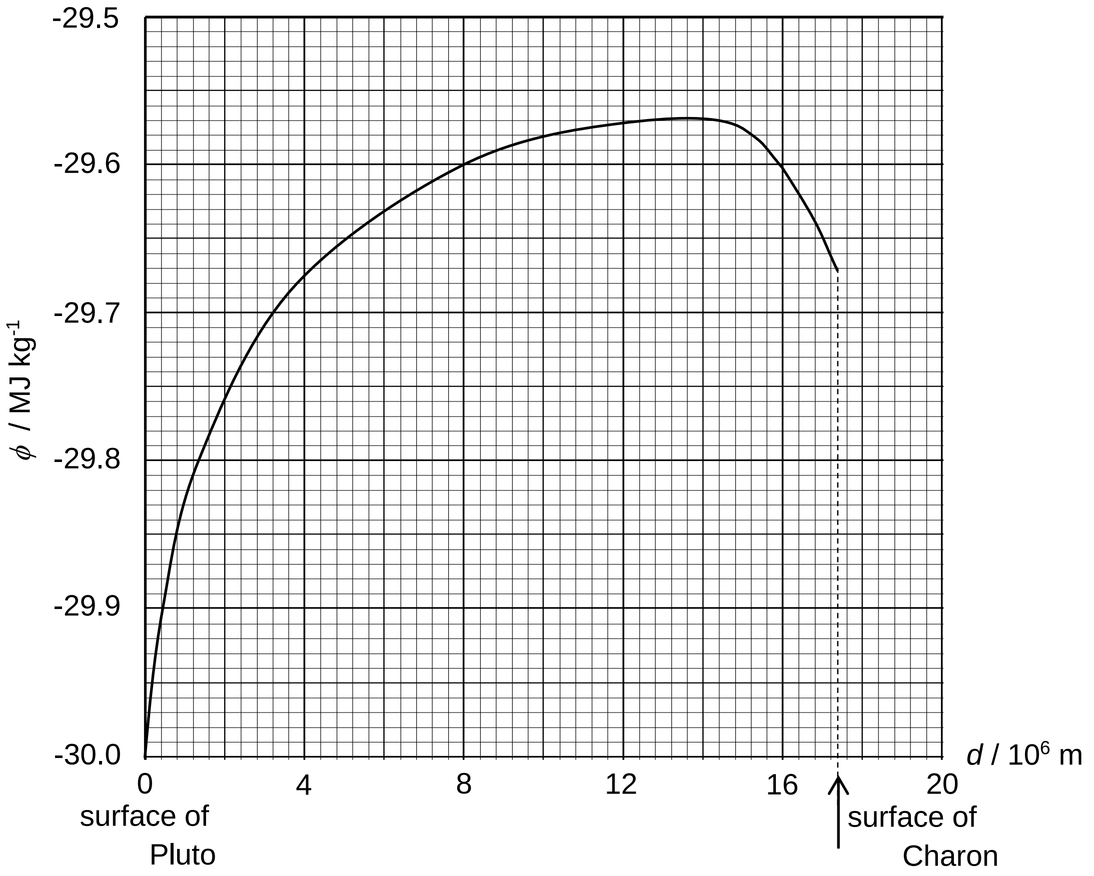

Fig. 8.1

The gravitational potential is taken as being zero at infinity.

(i) By considering the definition of gravitational potential, suggest why all values of ϕ in Fig. 8.1 are negative and explain how they may be obtained.

The direction of the external force to bring a (test) mass from infinity to a point in the field without a change in kinetic energy is opposite in direction to displacement. [B1]

Hence the work done per unit mass (which is the gravitational potential) is negative.

The resultant gravitational potential at that point is the (numerical) sum of the individual negative values of gravitational potential due to each planet. [B1] [2]

(ii) Explain how Fig. 8.1 can be used to determine the resultant gravitational force acting on an object with mass m placed at a point between Pluto and Charon.

The resultant gravitational force acting on an object at a point is the product of m [B1] and the (negative of the) gradient of the tangent at that point [B1] on the graph.

.... [2]

(iii) A lump of rock of mass 5.0 kg is ejected from the surface of Charon such that it travels towards Pluto.

1. Using data from Fig. 8.1, determine the minimum speed of the rock as it hits the surface of Pluto.

Loss in gravitational potential energy = gain in kinetic energy

$$m(\phi_i - \phi_f) = \tfrac{1}{2} m v_{minimum}^{2} - 0 \quad \text{[M1]}$$

$$5.0\,(-29.57 - (-30.00)) \times 10^{6} = \tfrac{1}{2}(5.0) \times v_{minimum}^{2} \quad \text{[C1]}$$

$$v_{minimum} = 927 \text{ m s}^{-1} \quad \text{[A1]}$$

minimum speed = ... m s^{-1} [3]

2. The rock is now projected from Pluto to Charon.

State and explain how its minimum speed when it hits the surface of Charon will be different from the answer in **(a)(iii)1**.

The loss in gravitational potential energy from the maximum gravitational potential energy point to the surface of Charon is less than that to the surface of Pluto. Thus, the gain in kinetic energy of the rock on impact on Charon will be lesser [M1] and the minimum speed of impact on Charon will be smaller. [A1]

.... [2]

(b) In another isolated planetary system, X and Y are two stars of equal mass M and separated by a distance $2L$. The two stars rotate about their common centre of mass.

(i) State Newton's Law of Gravitation.

The gravitational force of attraction between any two point masses is directly proportional to the product of their masses and inversely proportional to the square of their separation. [B1] [1]

(ii) Explain why the two stars will not collide with each other even though the gravitation force acting on each other are attractive.

The gravitational force acts perpendicular to the motion of the star. [B1] Hence the gravitation force provides the centripetal force for circular motion. [B1]

.... [2]

(iii) Derive an expression, in terms of M, L and the gravitational constant G, for the period T of their rotation.

Gravitational force provides the centripetal force for circular motion.

$$\frac{GMM}{(2L)^2} = ML\left(\frac{2\pi}{T}\right)^2 \quad \text{[B1]}$$

$$T = \sqrt{\frac{16\pi^2 L^3}{GM}} \quad \text{[B1]}$$

[2]

(c) Fig. 8.2 shows three points A, B and C relative to the stars X and Y. All three points are equidistant from both X and Y and point B lies on the straight line joining the centres of mass of X and Y.

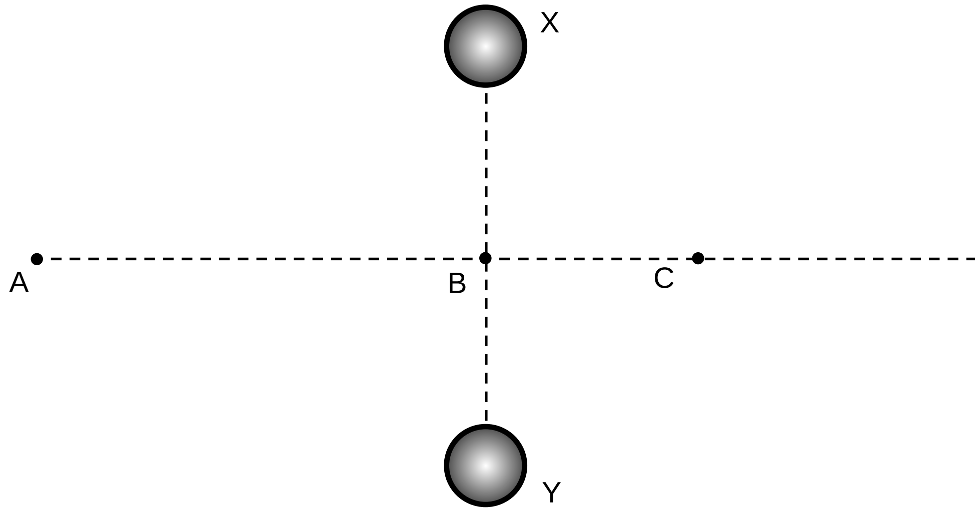

Fig. 8.2

(i) Draw arrows on Fig. 8.2 to represent the gravitational forces due to each star acting on a small mass placed separately at points A, B and C. The length of the arrow should be relative to their respective magnitudes. [2]

arrows directed towards centres of X and Y [B1]
shortest arrows for A and largest for B [B1]

(ii) Fig. 8.3 shows another schematic of X and Y. P is a point equidistant from X and Y and is at a distance *s* from the straight line joining the centers of mass of X and Y.

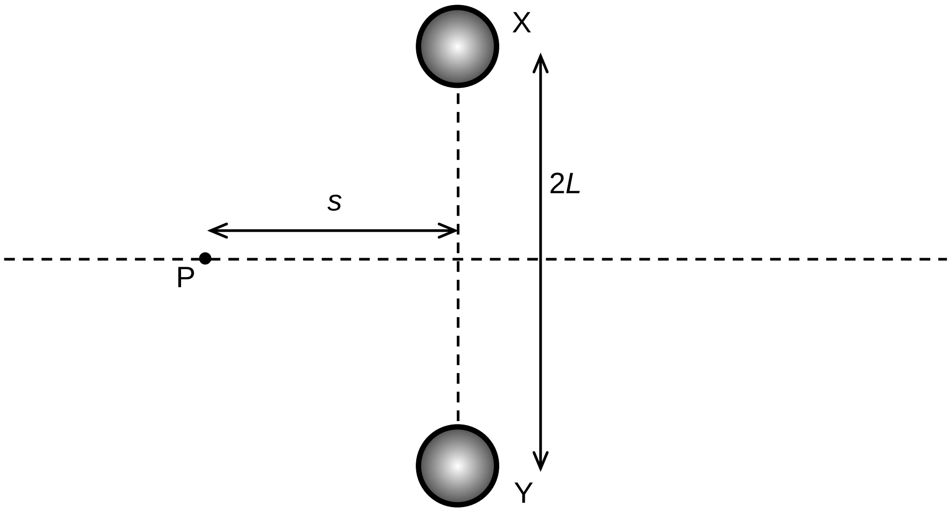

Fig. 8.3

1. Show that the magnitude of the resultant gravitational force acting on a mass *m* placed at P is

$$F_G = \frac{2GMms}{\left(s^2 + L^2\right)^{\frac{3}{2}}}$$

Components of forces along XY cancels each other [B1]

$$F_G = 2\frac{GMm}{\left(\sqrt{L^2 + s^2}\right)^2}\left(\frac{s}{\sqrt{L^2 + s^2}}\right) \text{ [B1]}$$

$$= \frac{2GMms}{\left(s^2 + L^2\right)^{\frac{3}{2}}} \text{ (shown)}$$

[2]

2. Hence, state and explain if the subsequent motion of a mass placed at rest at point P is simple harmonic.

From the above equation, it may be deduced that the acceleration of the mass is not directly proportional to its displacement from equilibrium *s* [M1], the subsequent motion of the mass is not simple harmonic [B1]
.... [2]

9 **(a)** One assumption of the kinetic theory of gases is that the particles of the gas make perfectly